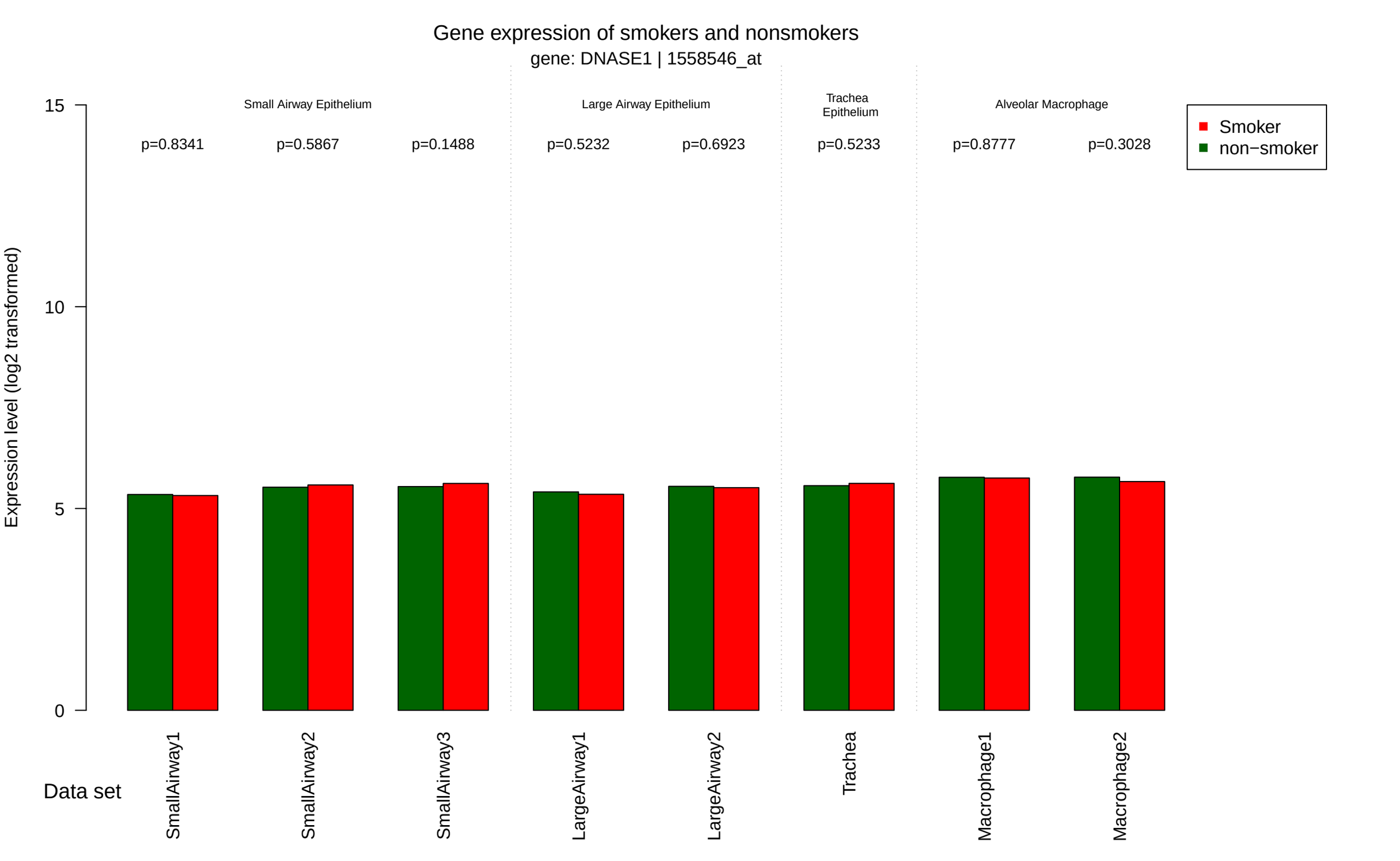
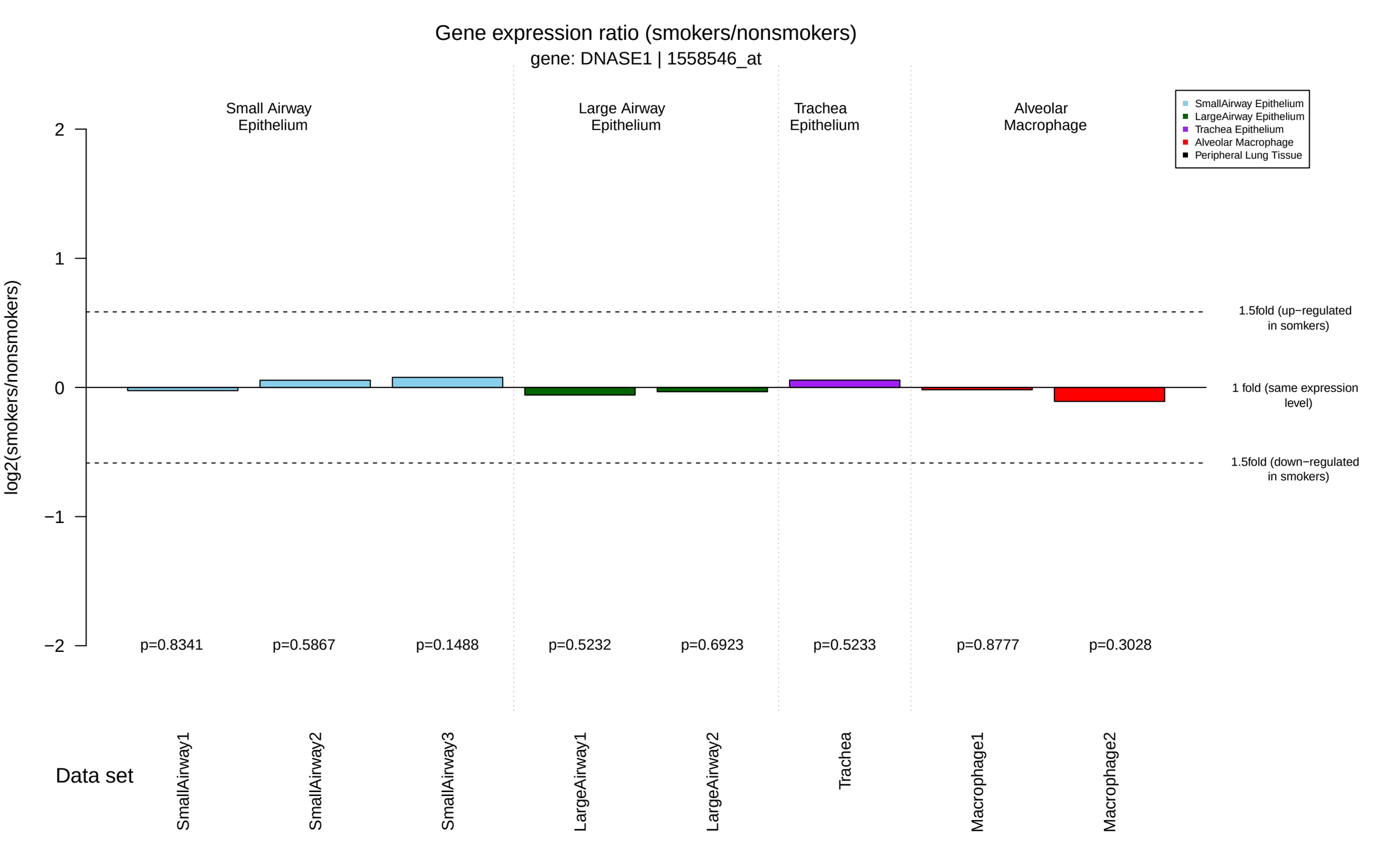
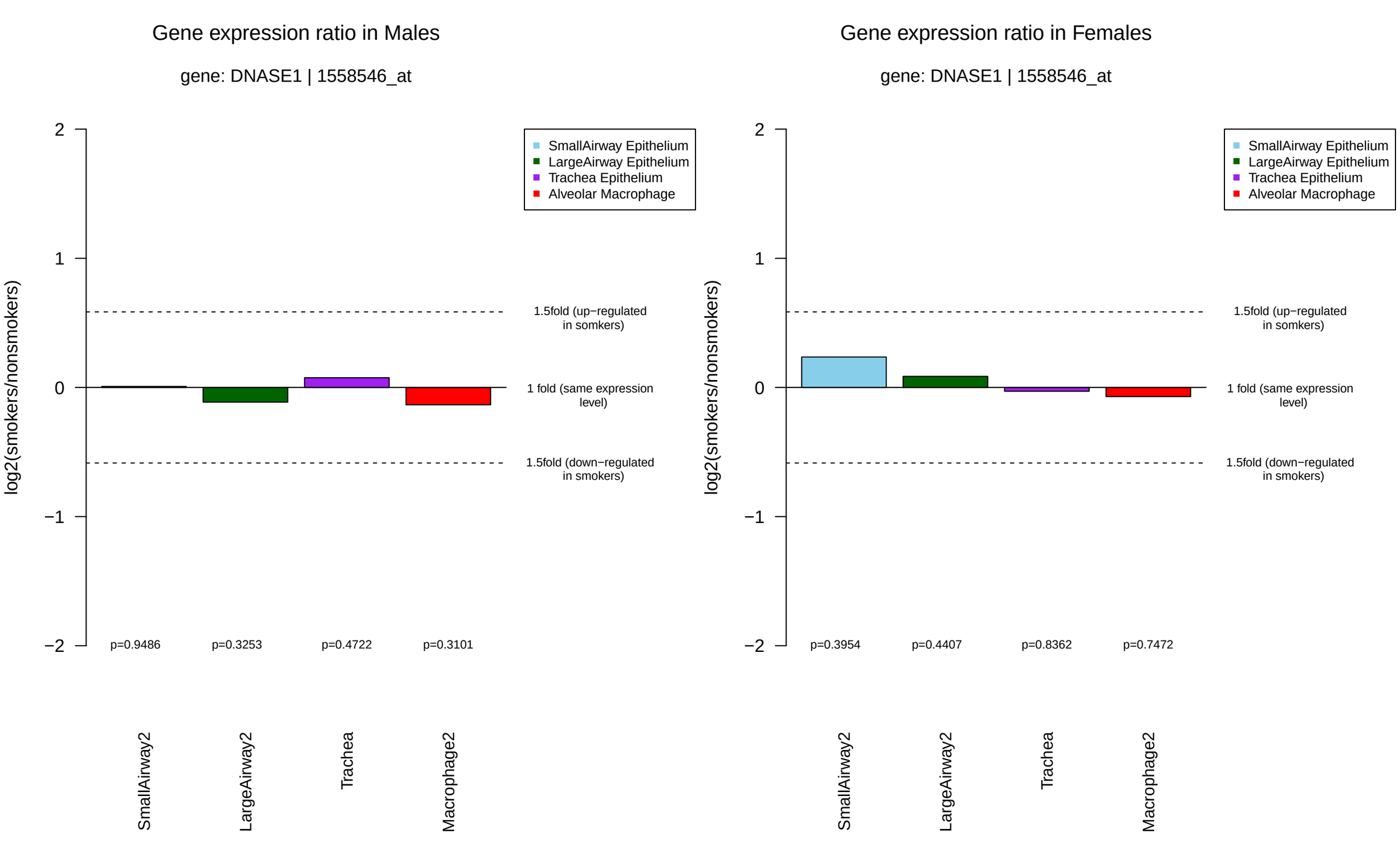
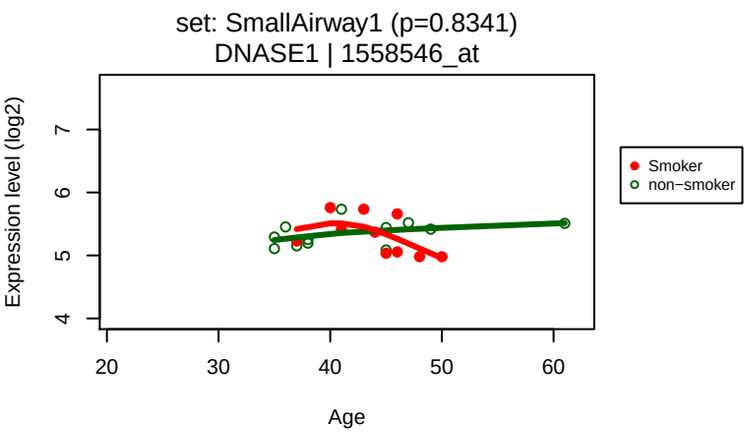
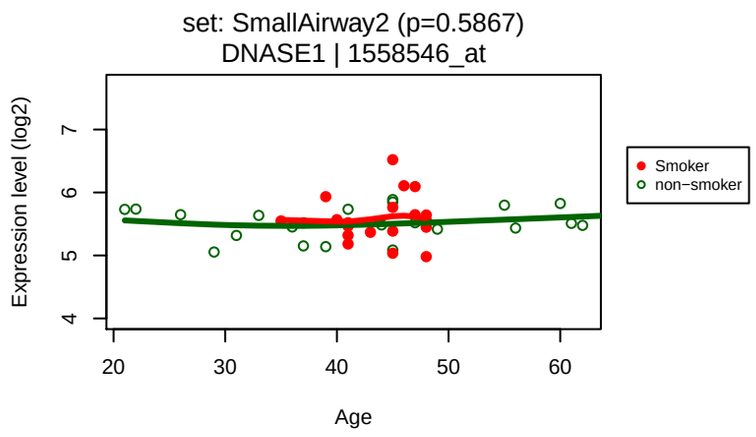
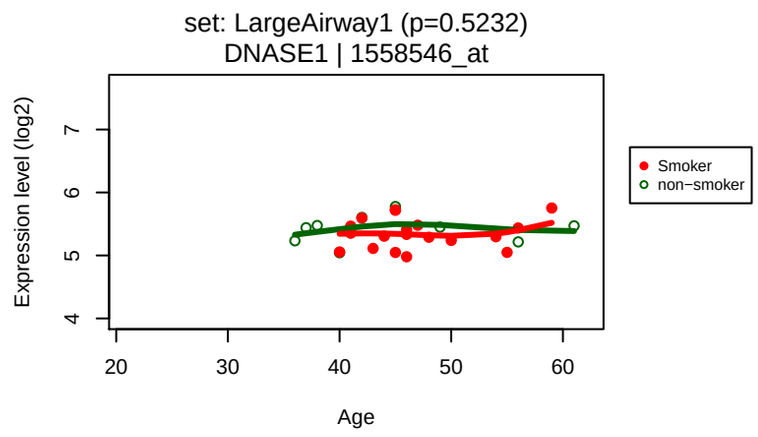

Gene expression of smokers and nonsmokers
gene: DNASE1 | 1558546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8341
p=0.5867
p=0.1488
p=0.5232
p=0.6923
p=0.5233
p=0.8777
p=0.3028
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

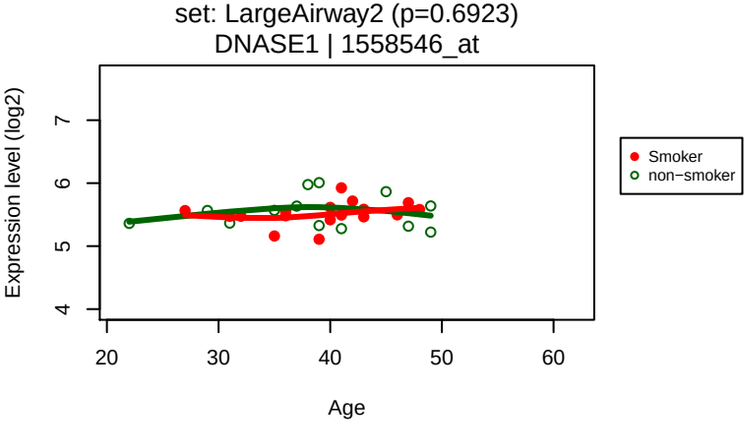

Gene expression ratio (smokers/nonsmokers)
gene: DNASE1 | 1558546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8341
p=0.5867
p=0.1488
p=0.5232
p=0.6923
p=0.5233
p=0.8777
p=0.3028
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

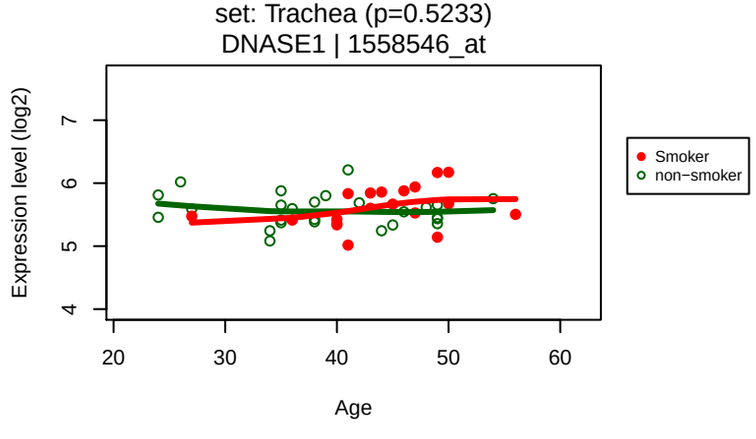

Gene expression ratio in Males
gene: DNASE1 | 1558546_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.9486
p=0.3253
p=0.4722
p=0.3101
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: DNASE1 | 1558546_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3954
p=0.4407
p=0.8362
p=0.7472
SmallAirway2
LargeAirway2
Trachea
Macrophage2

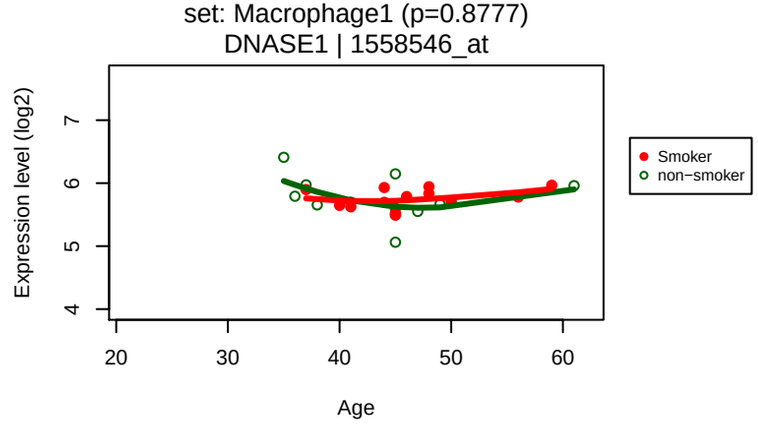

set: SmallAirway1 (p=0.8341)
DNASE1 | 1558546_at
Expression level (log2)
Age
Smoker
non−smoker

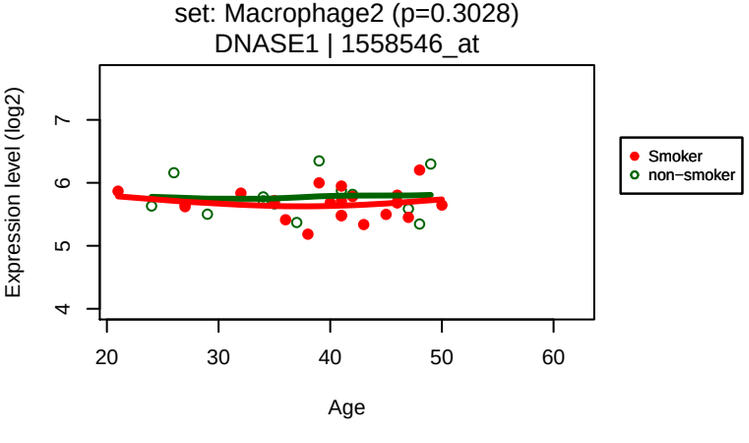

set: SmallAirway2 (p=0.5867)
DNASE1 | 1558546_at
Expression level (log2)
Age
Smoker
non−smoker

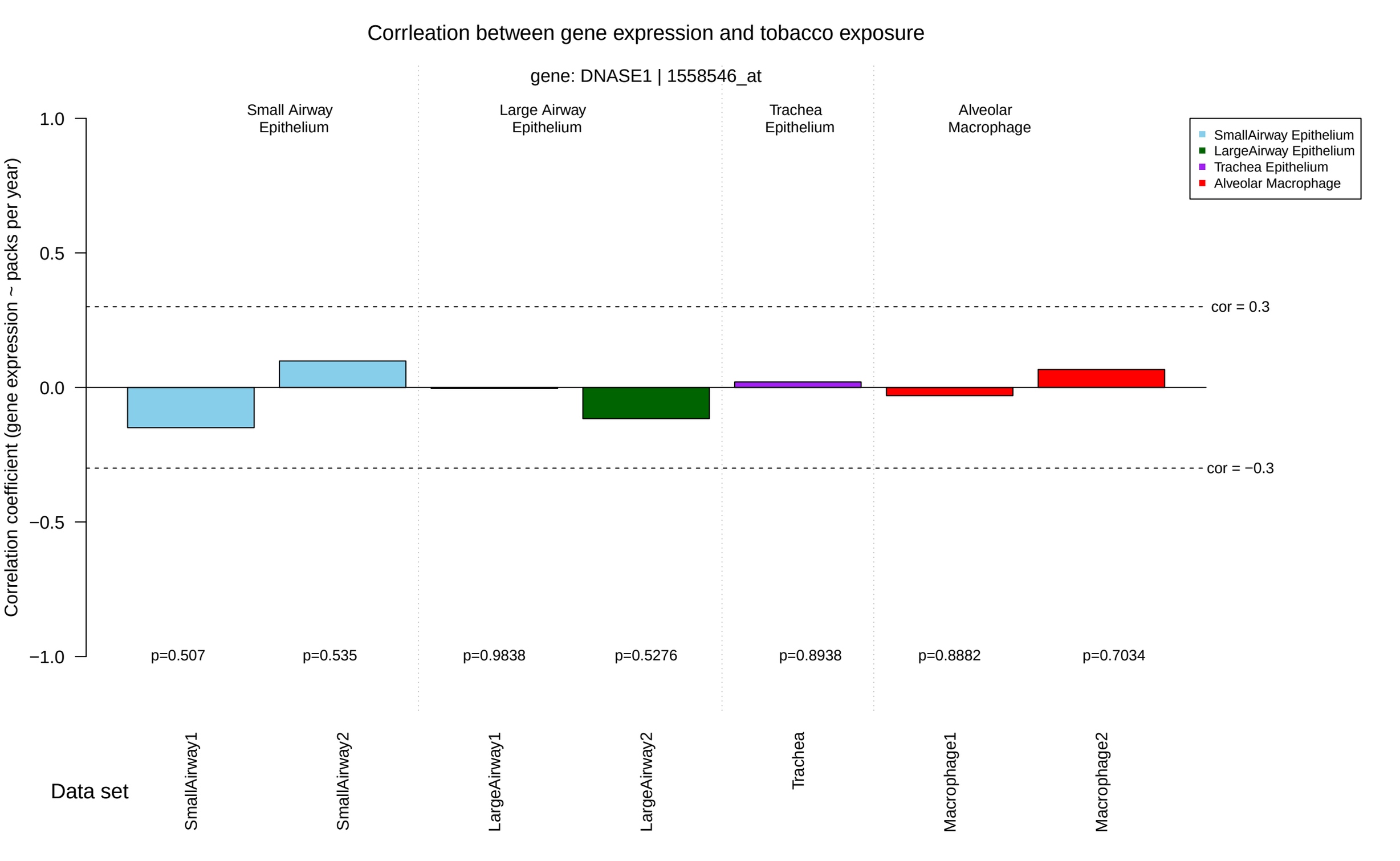

set: LargeAirway1 (p=0.5232)
DNASE1 | 1558546_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.6923)
DNASE1 | 1558546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.5233)
DNASE1 | 1558546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.8777)
DNASE1 | 1558546_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.3028)
DNASE1 | 1558546_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: DNASE1 | 1558546_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.507
p=0.535
p=0.9838
p=0.5276
p=0.8938
p=0.8882
p=0.7034
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2